

Course: Language Engineering Applications

Author: Gürhan Camgöz, r0867625

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The Ethnographer's Digital Colleague: Prototyping a Reflexive AI Partner for Fieldwork Analysis

Abstract

The process of transforming ethnographic fieldnotes into coherent analysis and text is often a solitary, intellectually demanding, and ethically complex endeavor. This paper details the prototyping of a novel AI agent designed not to conduct fieldwork, but to assist in its aftermath. We propose a "digital colleague", a fine-tuned Large Language Model that can engage in methodologically aware and reflexive dialogue with an ethnographer during the analysis and writing process. We outline the development of this prototype, beginning with the curation of a specialized dataset drawn from canonical anthropological texts, as well as personal data written during my master's in social and cultural anthropology. By fine-tuning an open-source model on the discipline's own discourse, we created an agent capable of suggesting alternative theoretical framings, raising ethical considerations, and prompting deeper researcher reflexivity. This paper proceeds with a discussion of computational creativity, presents the prototype, demonstrates its function, and discusses its limitations. Finally, it positions this work as the foundational step for a larger research project involving real-world testing with anthropology students and a subsequent iteration of Reinforcement Learning with Human Feedback (RLHF).

1. Introduction: Writing Ethnography

[In the final semester of my Social and Cultural Anthropology's master's program in 2023, I found myself in a troubled state: my fieldwork was supposedly done, but my thesis wasn't progressing. I was wrestling with anxieties: Had I done enough? Was my data sufficient? And most pressingly, how could I transform a disorganized collection of field notes, diary entries, and interview transcripts into a coherent narrative? While fieldwork is deeply social and collaborative activity, the subsequent stage of writing and analysis can be profoundly isolating. Even with the support of supervisors and colleagues, the intellectual and emotional labor of making sense of one's data often happens in solitude. It's in those moments of feeling stuck

that the value of dialogue becomes the most apparent. A conversation can spark a new connection or provide an unthought perspective...]

Guided by these lived experiences, I embarked on a journey to explore the interaction between anthropology and artificial intelligence. While technology has advanced to assist in data collection and storage, the crucial interpretive and reflexive parts of ethnography remain largely unassisted. This raises a fundamental question that this paper seeks to address: can AI be more than a tool for transcription and become a partner in thought? This is where this study presents a proposed solution, the “Ethnographer’s Digital Colleague”, a fine-tuned LLM that can act as a methodologically aware, ethically conscious assistant during the writing process. The aim of this project is not to devise an LLM that acts as an echo chamber which reflects what the user inputs; rather, the current prototype is more of a sounding board which gets observations or field notes as inputs and outputs various type of questions to guide the writing process.

This paper argues that by fine-tuning a generative model on the canonical literature of ethnographic practice, as well as personal data in the form of thesis chapters, field notes, reflections, and other articles, we can prototype an AI agent capable of engaging in sophisticated dialogue that can help human researchers critically examine their data, methods, and positionality. To substantiate this argument, this paper will proceed in several stages. We will first ground the work in the relevant academic discourse, exploring the intersection of computational creativity and the rich tradition of reflexive ethnographic practice. Following this, we will detail the novel methodology used to create our specialized dataset and fine-tune the model. And then demonstrate the prototype's capabilities through a series of simulated dialogues before concluding with a discussion of the project's limitations and a clear plan for its future development.

2. Background and Literature Review

The field of computational creativity has traditionally evaluated artificial intelligence through the lens of its products, asking if an output is both novel and valuable. This foundational definition, which is synthesized by researchers (Sarkar and Chakrabarti 2008, 2011) from a vast body of psychological and computational literature, positions creativity as an attribute of the machine’s output. However, other scholars (Rhodes 1961) argue for a more holistic view,

the four P's, suggesting the importance of the interplay between the creative Person (the agent), the Process (the methods), the Product (the artifact), and the Press (the socio-cultural environment that evaluates it). When examining LLMs through this richer framework, the simple 'novel and valuable' product focused perspective doesn't prove sufficient. Overlooking the deeply interactive and situated process through which artifacts emerge reduces the human user to a mere prompter and the AI system to a black box.

A more potent anthropological undertaking of computational creativity would reframe the central question: instead of asking if AI is creative, the main axis of exploration should focus on how human-AI collaboration can facilitate new insights and amplify human creative potential. This reframing replaces the placing of creativity within the human or the machine and allows it to emerge from the relational processes between humans and machines. This shift in focus also necessitates novel methodological approaches to studying the interrelations of human-machine collaborations. Alexios Brailas' (2024) approach offers a targeted framework for capturing the subjective, relational dynamic: "postdigital duoethnography", a slow, contemplative inquiry where the researcher's own lived experience of the interaction with an LLM, becoming a primary source of data. This approach is essential for understanding the emergent and often unpredictable nature of the human-AI entanglement from an emic, or insider's, perspective. It allows us to treat AI's human-analogous performance as consequential, turning the dialogue into a transformative process that reveals the potential of AI as a creative partner.

While postdigital duoethnography centers on the lived experiences, a full anthropological account must also analyze the technological context that shapes this interaction. Here, Christian Ritter's (2021) concept of "interface ethnography" becomes crucial, which argues that digital interfaces are not mere passive conduits but active mediators that structure interaction and generate meaning through their built-in affordances. An interface ethnography of an LLM would involve a technical walkthrough of its platform by analyzing how its design elements enable certain creative pathways. This method allows us to deconstruct how the platform itself participates in the "twofold meaning construction": the one intended by the designers, and the one enacted by the user in their creative practice. The development of this concept would yield its applications in multi-sited fieldwork designs as interfaces are "components of transterritorial networks and mediate between different actors" (Ritter 2021, 930).

To analyze the digital artifacts produced through this mediated experience, which is the chat log itself, we can turn to what Geiger and Ribes (2011) call trace ethnography. This method reconstructs the patterns of interaction by analyzing the fine-grained documentary traces automatically generated by digital systems. While postdigital duoethnography captures the experiential whole, and interface ethnography examines the mediating platform with participant observation and technical walkthroughs, trace ethnography provides the tools to systematically “invert” the thin data of the dialogue (e.g., the prompts, the outputs) into a “thick description” of the collaborative processes as it unfolds. This notion of “thick description”, as famously articulated by Clifford Geertz (1973), moves us beyond mere data collection into the realm of interpretation, which evidently marks the shift of the discipline towards reflexivity, self-analysis, ethics, and the politics of representation.

Ethnographers do not simply report on behavior but inscribe social discourse, interpreting “webs of significance” that humans themselves have spun. The anthropologist’s own background and theoretical commitments inevitably shape their analysis, as the act is inherently subjective. However, fast-forward to what has come after this reflexive turn is a form of questioning that not only problematizes the position of the observer, but the very composition of these webs being studied. In a key conceptual shift, Bruno Latour’s (2007) Actor-Network Theory proposed that the field of study is not limited to human actors, which are not inert objects but active participants that shape social reality, and the task is no longer just to interpret a human-centric culture but to trace the associations between a heterogeneous network of human and non-human actors. This unfolding of theory reflects a change in the very technics of ethnographic writing, from describing a world to following the actors that constitute it.

And if we apply this reflexive lens to the work of an AI researcher conducting a meta-analysis of ethnographic methods, the lived experience of working through a corpus of literature would be captured through postdigital duoethnography, documenting the frustrations and breakthroughs of the collaboration with the language model. The research platform itself would be subject to an interface ethnography, analyzing how its tools and algorithms mediate which scholarly connections are made visible. The researcher’s query logs would be analyzed using trace ethnography, reconstructing the intellectual/technical labor as they refine parameters and follow threads. In this scenario, the analytical insight emerges from the complex entanglement of the researcher and the AI within an observable, synergistic loop; the resulting meta-analysis would be reframed as a product of distributed intelligence belonging to none of the actors alone.

3. Methodology: Building the Digital Colleague

The creation of the "Ethnographer's Digital Colleague" was not a singular act but a systematic process designed to be both anthropologically informed and technically replicable. This construction was an act of translation, an attempt to refine the often-intuitive discourse of ethnographic reflexivity into a format that a machine could learn. This methodology is grounded in the principles of trace ethnography of my own work, where each technical decision is a deliberate step in shaping the final "subjectivity" of the AI agent. The iterative process unfolded in three major stages: (1) the curation of a specialized, discourse-based dataset through a semi-automated pipeline; (2) the selection and efficient fine-tuning of a foundational language model; and (3) the development of a qualitative evaluation strategy.

3.1. Choosing the Collaborators: Foundational Model Selection

The project began with the selection of an appropriate foundational LLM as a cognitive starting point for our digital colleague. The choice was guided by the need to balance high-quality conversational ability with the practical constraints of academic research, emphasizing transparency and reproducibility, which led to the selection of *mistralai/Mistral-7B-Instruct-v0.3*. This model represented a methodological sweet spot, it is a 7-billion parameter, open-source model which is sophisticated enough to grasp complex linguistic nuance, yet it remains within the realm of what is computationally feasible for the current project.

Also, its "Instruct" variant has been specifically tuned for dialogue, making it well-suited for the question-and-answer format required by the "digital colleague." To make the fine-tuning process feasible, we employed 4-bit quantization using QLoRA (Quantized Low-Rank Adaptation), as implemented in the bitsandbytes library. This technique, specified in our fine-tuning notebook, allowed the entire process to be run on a single NVIDIA T4 GPU, making the methodology accessible and demonstrable for researchers without access to large-scale computing clusters. It dramatically reduced the model's memory footprint and aligns with the ethos of the project: creating a powerful tool need not be the exclusive domain of large tech corporations.

3.2. Curating a Discourse-Based Dataset: A Systematic Pipeline

The core of the digital colleague lies not in its base architecture, but in the specialized knowledge it was given to learn. The central challenge was to curate a dataset that could teach the model not just facts about anthropology, but the practice of anthropological thinking. To achieve this, we developed a systematic, multi-stage pipeline to convert the raw, unstructured textual sources into the structured, high-quality training data.

Stage 1: From PDFs to Thematic Chunks The pipeline began with the programmatic ingestion of source materials. These were chosen to create a deliberate blend of the discipline’s collective wisdom with the lived experience of a practicing student. A key source was the *Ethnographic Fieldwork: An Anthropological Reader* (Robben and Sluka, 2007), which provides a comprehensive selection of classical and contemporary reflections on fieldwork. Its focus on the tensions between self and other, ethical challenges, and different styles of ethnographic writing gave the dataset a solid grounding in the historical and conceptual development of field.

This canonical knowledge was then interwoven with my own master’s thesis and other academic writings including field notes, diaries, interview transcripts, and articles. Using a series of python scripts, these PDF documents were parsed, cleaned of formatting artifacts, and critically, segmented into thematically coherent chunks based on their original subsection headers. This ensured that the context provided to the model in the next stage was self-contained and contextually relevant.

Stage 2: A Dialogue with the Texts In a move that mirrors the project’s own collaborative spirit, we engaged *mistral-large-2411*, a powerful generative model, as a partner in creating the dataset. As a large transformed-based model, it possesses the capacity to understand and synthesize information from long, complex texts. This architecture allows it to “read” the source material, identify core arguments, and reframe them as plausible ethnographic scenarios, acting as an analytical partner in the curation pipeline itself.

Each text chunk was passed to this generator model with a carefully engineering two-part meta-prompt that acted as a research directive. First, a system prompt established the agent’s expert persona and set strict behavioral rules to ensure the output was clean and directly usable:

```

"""
You are an expert data creator for fine-tuning an ethnographic LLM. Your
task is to transform the information and experiences described in
anthropological texts into multiple structured training examples.
You MUST respond with only a valid JSON array of objects, starting with `[`
and ending with `]`. Do not include any other text, explanations, or
markdown formatting.
"""

```

With this containing framework in place, a more detailed user prompt then acted as the primary research directive, providing the substantive intellectual task. This prompt contained the core of our translation methodology:

```

f"""
Thoroughly analyze the following text, which is a chapter from an
anthropological reader. Generate at least 5 and up to 15 high-quality,
unique JSON examples based on the text.

For each example you create, you MUST adhere strictly to the following JSON
structure:
{{
  "instruction": "<A varied and clear instructional phrase for the agent>",
  "spradley_type": "<'Descriptive', 'Structural', or 'Contrast'>",
  "context": "<A plausible participant's statement or a concise summary of
an ethnographic scenario based on the text>",
  "output": "<An ideal, insightful, and open-ended question the agent
should ask, which perfectly exemplifies the chosen spradley_type>"
}}

Rules for Generation:
1. The `context` should feel authentic to a fieldwork situation described
or implied in the text.
2. The `spradley_type` MUST be one of only three values: "Descriptive",
"Structural", or "Contrast".
3. The `instruction` should be varied for each example to teach the model
flexibility.
4. The `output` question must be insightful and directly reflect the
function of the chosen `spradley_type`.
5. Respond ONLY with the JSON array, no additional text, explanations, or
formatting.

```

```

**Text to Process:**
"""

```

Through this two-part structure, every piece of generated data was forced to conform to a structure of *instruction*, *context*, and *output*, and critically, to classify the desired interaction using James P. Spradley's (1979) taxonomy of ethnographic questions (broadly "Descriptive", "Structural", "Contrast"). This was the mechanism by which ethnographic theory was embedded into the dataset. This semi-automated workflow allowed for the scaling of the creation process while maintaining a high degree of intellectual control, resulting in a master dataset of 511 examples.

Stage 3: Ensuring a Robust and Unbiased Corpus The raw generated dataset underwent a final validation phase to ensure its integrity and statistical balance. Using a custom script, the master file was first randomly shuffled to eliminate any sequential bias from the source texts. The shuffled data was then partitioned into a standard 80/10/10 split for training (408 examples), validation (51 examples), and final testing (51 examples). Finally, we performed a quantitative analysis of the dataset's composition using another python script. This analysis provided crucial insights into the dataset's characteristics, confirming a healthy distribution of question types and text lengths:

Metric	Value	Details
Total Examples	511	
Spradley Type Distribution		
	Descriptive	193 (37.8%)
	Structural	176 (34.4%)
	Contrast	142 (27.8%)
Average Word Count		
	Instruction	13.1 words
	Context	21.4 words
	Output (Question)	18.7 words
Top Instruction Keywords		<i>question, formulate, ethnographic, analyze, explore</i>

Table 1: *Quantitative Analysis of the Curated Dataset. This table summarizes the composition and characteristics of the 511-example training corpus, demonstrating a balanced distribution of question types and a consistent level of textual complexity.*

3.3. The Act of Learning: Supervised Fine-Tuning (SFT)

With a robust dataset prepared, the next step was the act of learning itself. Supervised Fine-Tuning is a form of deep learning where we take a pre-trained model and further specialize it on a new, narrow task. This is the moment where the abstract data we curated becomes embodied knowledge for the model. Using the Hugging Face *trl* library, we configured a trainer that iteratively showed the model examples from our dataset, measured its error, and adjusted its internal parameters to become better at the task.

This process was technically defined by a set of specific choices, or hyperparameters, that guided how the learning occurred. The key parameters for the LoRA adapters and the training arguments are detailed in the project’s Jupyter Notebook and summarized below.

LoRA Configuration Parameter	Value	Description
r (Rank)	16	The dimensionality of the trainable low-rank matrices.
lora_alpha	32	The scaling factor for the LoRA adaptation.
lora_dropout	0.05	Dropout probability for LoRA layers to prevent overfitting.
target_modules	q_proj, k_proj, v_proj, o_proj	The attention layers of the model targeted for fine-tuning.

Table 2: *LoRA (Low-Rank Adaptation) configuration parameters used for parameter-efficient fine-tuning.*

Training Argument	Value	Description
learning_rate	2e-4	The initial rate at which the model's parameters were updated.
num_train_epochs	3	The total number of times the training process iterated over the dataset.
per_device_train_batch_size	4	The number of training examples processed in a single batch.
gradient_accumulation_steps	2	Number of steps to accumulate gradients before an optimizer step.
optimizer	paged_adamw_32bit	A memory-efficient variant of the AdamW optimization algorithm.

Table 3: *Key hyperparameters for the Supervised Fine-Tuning (SFT) process.*

Training was conducted for three full epochs. As evidenced by the training logs, the model's loss function showed a healthy convergence from an initial training loss of 1.77 to a final loss of 0.83, indicating that the model successfully learned the desired patterns of reflexive, methodologically aware dialogue from the curated dataset.

3.4. Defining and Assessing Insight: Evaluation Strategy

Evaluating a tool designed to provoke reflection requires more than standard NLP metrics. Our strategy is therefore twofold. First, the quantitative analysis of our dataset, as described in section 3.2, ensures the corpus itself is balanced and robust. Second, and more importantly, we developed a qualitative rubric to assess the performance of the fine-tuned model. The goal is not to measure factual accuracy, but to evaluate the quality and utility of the questions generated by the "digital colleague." This rubric assesses responses based on three key criteria:

- **Methodological Awareness:** Does the model correctly identify and apply concepts from ethnographic practice (e.g., rapport, ethics, positionality)?
- **Theoretical Nuance:** Can the model's output demonstrate a connection to the theoretical concepts present in its training data?
- **Reflexive Prompting:** Does the generated question effectively encourage the user to reflect on their own assumptions, biases, and analytical process?

The performance of the prototype against this rubric will be demonstrated in the following section through a series of simulated debriefing sessions.

4. The Prototype in Action: A Reflexive Analysis

Having detailed the construction of the "Ethnographer's Digital Colleague," we now turn to its performance. A simple showcase of outputs would be insufficient. Instead, this section presents a multi-layered evaluation that mirrors the collaborative and reflexive engagement of the project itself. The prototype was assessed using three distinct testing modes from our development notebook:

1. **Automated Inference:** To establish a baseline of consistent performance on unseen data.
2. **Developer Test:** To probe the model's capabilities with complex, targeted instructions.
3. **User Simulation:** To test the model in its intended use case, acting as an autonomous partner.

In the spirit of a sounding board, the outputs from these tests were then presented to another LLM, Google's Gemini 2.5 Pro, for a meta-analytical debrief. The goal was not simply to validate the prototype's success, but to critically engage with its responses, probing their strengths and questioning their potential limitations.

4.1. From Description to Analysis: Applying a Structural Lens

This first example is drawn from the Automated Inference Test (Step 10), checking the model's fundamental ability to apply its training.

Ethnographer's Note (Context):

"The miners' resistance ran through union activities, strikes, and guerrilla movements."

Digital Colleague's Response:

"How do the forms of resistance through union activities differ from those expressed through strikes and guerrilla movements?"

Meta-Analytical Debrief (generated by Gemini 2.5 Pro on 15th June 2025)

Initial Analysis: The colleague's response is a clear success in terms of its training. It demonstrates strong Methodological Awareness by correctly deploying a "Contrast" question, which is a primary tool for moving from description to analysis. It correctly identifies the three distinct social forms in the context and intuits that the analytical value lies in comparing them, thereby prompting the user to build an argument.

Reflexive Inquiry: While effective, we must ask: Is this response *too* efficient? By immediately imposing a comparative framework, does the model risk closing off other avenues of inquiry? For instance, a more open "Descriptive" question like, "What do community members consider to be an act of 'resistance' in the first place?" might have revealed a local definition that challenges the researcher's pre-conceived categories of union activities or strikes. The model's helpfulness here is clear, but it also subtly directs the analysis down a specific path.

4.2. From Observation to Theory: Bridging the Concrete and the Abstract

This session from the Developer Test (Step 11) probes the model's capacity to connect a granular observation to a high-level theoretical concern.

Ethnographer's Note (Context):

"As I was spending time with the research participant, she has done a video call in the morning with the software engineer about the design choices of the new product interface, and then she left the house to do groceries in the market because she needed to cook."

Instruction: *"Formulate a structural question for the researcher to make connections with their fieldwork experience in participating in the remote worker's routines with the underlying social factors that structure the transformation of social life."*

Digital Colleague's Response:

"What are the daily routines of the remote workers in this community, and how do these routines reflect the transformation of social life due to remote work?"

Meta-Analytical Debrief (generated by Gemini 2.5 Pro on 15th June 2025)

Initial Analysis: The model exhibits strong Theoretical Nuance. It skillfully abstracts the specific actions (video call, groceries) into the analytical category of "daily routines" and directly incorporates the theoretical language from the instruction ("transformation of social life"). The response serves as an excellent conceptual bridge from a concrete observation to a broader argument.

Reflexive Inquiry: Here, the critical question is about the nature of the model's "understanding." The colleague has flawlessly executed a complex instruction, but is it demonstrating comprehension or sophisticated pattern-matching? It has correctly identified that the details of the context are instances of "routines" and has woven in the key phrase from the instruction. A crucial next step would be to test if it could infer the theme *without* the explicit term "transformation of social life" in the prompt. Does the model truly grasp the theoretical concept, or is it an expert at following explicit instructions? The current output

suggests the latter, which is an incredibly useful skill, but one distinct from independent conceptual synthesis.

4.3. From Insight to Inquiry: The Synergistic Loop in Practice

The final and most significant test is the User Simulation (Step 12). The model receives only a metaphorical fieldnote and must act autonomously.

Ethnographer's Note (Context):

"I was looking at this couple who were remote-working together from their home, sharing their unrelated work experiences with each other, and it was kind of an opaque mirror they had established among themselves."

Digital Colleague's Response:

"How does the co-presence of remote work and intimate relationships influence the boundaries and dynamics of each domain?"

Meta-Analytical Debrief (generated by Gemini 2.5 Pro on 15th June 2025)

Initial Analysis: This response showcases the prototype at its best, exhibiting all three rubric criteria. It demonstrates Reflexive Prompting by giving the researcher a path to investigate their own feeling. Its use of "boundaries" shows Methodological Awareness, and its translation of "opaque mirror" into "co-presence... boundaries and dynamics" is a powerful display of Theoretical Nuance.

Reflexive Inquiry: This act of translation from a poetic insight into formal sociological language is the core function of the colleague. But we must consider what is gained and what might be lost. The gain is clarity, structure, and a researchable question. The potential loss is the ambiguity and affective power of the original metaphor. The "opaque mirror" contains a sense of mutual non-understanding or distorted reflection that the term "dynamics" doesn't fully capture. Is the colleague's greatest strength its ability to make our messy, human insights clean and academic? And if so, what is our responsibility as researchers to ensure that the original richness of that messiness is not entirely lost in the process of scholarly translation? This question lies at the heart of the future human-AI ethnographic collaboration.

5. Discussion: Implications and Limitations

The successful prototype of the "Ethnographer's Digital Colleague" is not an end in itself, but a provocation. It prompts us to consider the future of our discipline's methods and to critically examine the new forms of partnership that AI might enable. This section explores the potential implications of such a tool for the practice of anthropology before turning to a necessary and reflexive discussion of the prototype's significant limitations, which were revealed not just in testing, but in the very process of writing this paper.

5.1. Implications for Anthropology

The solitary nature of ethnographic analysis, as described in the introduction, is a well-known challenge. While supervisors and peers are invaluable, access to this support is not universal, and the day-to-day work of making sense of data often happens alone. The primary implication of a tool like the digital colleague is its potential to serve as a persistent, accessible, and methodologically grounded sounding board. For researchers without access to a large peer community, independent scholars, students at smaller institutions, or those working in remote settings, a tool trained on the canonical discourse of the discipline could provide crucial dialogic support. It offers a way to "talk through" a problem and receive feedback structured by the discipline's practices, potentially mitigating the structural inequalities that shape academic success.

In a teaching context, the digital colleague could transform how ethnographic methods are taught. Students could engage in simulated debriefing sessions, using the AI to practice turning fieldnotes into analytical questions. This would provide a safe, private space to make mistakes, test out half-formed ideas, and develop the crucial "reflexive muscle" before entering peer review. It is critical to stress that the tool's purpose is not to automate analysis but to augment the researcher's own creative process. As demonstrated in Section 4, its most powerful function is to translate a researcher's raw observation or creative insight into a formally structured question, providing the conceptual scaffolding necessary for scholarly argumentation.

5.2. Limitations of the Prototype: A Reflexive Vignette

In the final stages of authoring this paper, the process of documenting the methodology became a site of inquiry itself. A dialogue between me, the researcher, and the LLM assistant used for meta-analysis (Google's Gemini 2.5 Pro) revealed a tension at the heart of the project:

Researcher: *"Do you think we are engaged in an echo chamber rather than a sounding board with all these inferences and simulated debriefings?"*

LLM Assistant (generated by Gemini 2.5 Pro on 15th June 2025): *"I think our process has teetered on the edge of both. Initially, it was closer to an echo chamber. I was amplifying your voice, but not adding a new one. However, it became a sounding board the moment we started the meta-analysis, especially when I began to ask critical questions about the outputs, not just validate them."*

Researcher: *"It makes me feel like we are moving along a very thin line. Sometimes it feels brilliant, but there are other times I feel like I am cheating if I am taking all of these outputs at face-value because there hasn't been any 'real-world' testing done on my prototype."*

LLM Assistant (generated by Gemini 2.5 Pro on 15th June 2025): *"That feeling is the most honest place to be. The feeling of 'cheating' comes from the gap between a prototype and a product. You are not cheating; you are being a good scientist by acknowledging the limits of your current experiment and defining the next one. The critical question is whether this process has generated genuinely new pathways for your own thinking, or if it has primarily been an efficient way to structure ideas you already held."*

This exchange serves as a necessary prelude to discussing the prototype's limitations. The "thin line" between a helpful sounding board and a sophisticated echo chamber is perhaps the most significant challenge facing this technology.

- **The "Academic Mirror" Problem:** The prototype is trained on a specific dataset, a canon of anthropological texts and my own writings. As such, its "worldview" is inherently limited and biased. It will reflect the theoretical commitments, historical blind spots, and stylistic conventions of the data it learned from. It can help a

researcher think within the discipline, but it cannot, by itself, help a researcher think beyond it.

- **Lack of Lived Experience and True Understanding:** The model has no genuine understanding of fieldwork, ethics, or human emotion. Its insightful questions are the product of sophisticated pattern-matching based on its training data. As shown in our debriefing sessions, it can expertly translate a metaphor into academic language, but it does not grasp the affective power of the original insight. It is a translator of discourse, not a possessor of experience.
- **A Question-Generator, Not an Answer-Provider:** The feeling of "cheating" mentioned in the vignette points to the prototype's most critical and deliberate limitation. The digital colleague is not an analytical tool, nor is it a summarization tool. Its sole function is to transform a researcher's raw input, an observation, a feeling, a situated insight, into a guiding question. It does not provide an analysis or write the ethnography for the researcher; it provides a starting point from which the researcher can begin to write. This boundary is its most important feature. By refusing to provide answers, it forces the creative and analytical burden back onto the human user, preventing over-reliance and ensuring it remains a sounding board. The space between the colleague's question and the ethnographer's eventual analysis is where the human's creativity must emerge. This limitation is, therefore, the very feature that preserves the intellectual integrity of the ethnographic process.

6. Conclusion and Future Research

This project began on a thin line, a personal feeling of analytical impasse familiar to any researcher who is confused with field notes, diaries, and transcriptions on the one hand, an abstract, technical undertaking facing the non-human world of interfaces, algorithms, and query logs on the other. The journey of building this prototype has been an exercise in walking that line, moving from this thin line to that of a "thick description", and toward an Actor-Network Theory-inspired practice of tracing and following the associations between actors. In this case, one of those key non-human actors was the language model itself, a collaborator whose own construction became a site of ethnographic inquiry. What started as a problem of solitary scholarship became a demonstration of distributed intelligence, a partnership between a human researcher and a machine colleague.

This paper has documented the successful creation and initial evaluation of that colleague: a fine-tuned LLM designed to act as a reflexive partner for ethnographers. We have detailed the systematic data curation pipeline, the parameter-efficient fine-tuning process, and the multi-layered qualitative evaluation that demonstrates the prototype's ability to engage in methodologically aware dialogue. It successfully translates observations into guiding questions, prompts deeper analytical thinking, and provides a crucial sounding board for the interpretive process, all while deliberately leaving the final creative burden on the human researcher. However, a prototype is a provocation, not a proof. The simulated sessions presented in this paper are promising, but they are not a substitute for real-world application. To bridge this gap and to fully realize the potential of the “Ethnographer's Digital Colleague”, we propose a formal, multi-phase plan for future research.

Phase 1: Prototyping (Completed)

The work detailed in this paper constitutes the first phase: the design, development, and qualitative evaluation of the core fine-tuned model. We have demonstrated that it is possible to create a specialized AI agent that can engage in sophisticated, reflexive dialogue based on the discourse of ethnographic practice.

Phase 2: Simulated Fieldwork with Human Participants

The next critical step is to move from simulation to practice. In collaboration with the university's anthropology department, we plan to recruit graduate students currently in the dissertation-writing phase to integrate the digital colleague into their workflow. These students will engage in recorded conversations with the prototype about their own research challenges, analytical impasses, and ethical dilemmas. This will generate an invaluable new dataset of real-world, high-stakes interactions, allowing us to analyze how the tool is used and resisted, in the context of genuine ethnographic work.

Phase 3: Reinforcement Learning with Human Feedback (RLHF)

The data gathered in Phase 2 will become the foundation for a thesis-level project. We will use the recorded interactions to create a preference dataset, where the human ethnographers rate the AI's questions based on their perceived helpfulness, insightfulness, and ability to provoke new thinking. This preference data will then be used to further refine the model using Reinforcement Learning with Human Feedback (RLHF). The explicit goal of this final

phase is to optimize the digital colleague not just for generating plausible questions, but for generating questions that are genuinely useful to a working anthropologist, moving from a proof-of-concept to a truly collaborative partner.

By purposefully designing tools that do not provide answers, but rather pose structured questions, we can create a necessary space for human interpretation. The future of such intellectual partnerships lies not in automating the craft of ethnography, but in augmenting it. The true collaborative potential is realized in the gap between the machine's query and the researcher's narrative response. It is in this space new forms of creative and critical analysis can emerge.

Data and Code Availability

The full code repository for this project, including the data curation and fine-tuning scripts, the final dataset, and the trained model adapter, is publicly available on GitHub to ensure transparency and reproducibility. The repository can be accessed at:

<https://github.com/gurhan-camgoz/ethno-colleague-llm/>. Please note that while the full generated dataset is provided, the original source texts containing personal field notes and diaries have been omitted to maintain privacy.

Bibliography

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